TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	76.8 ± 0.60
Correlation Stability	88.8 ± 0.87
Distribution Consistency	87.4 ± 0.66
Entropy Preservation	59.0 ± 0.89
Mean Z Score Drift	0.4 ± 0.01
Categorical Fidelity	91.5 ± 5.00
Privacy Risk	36.7 ± 4.03
Data Density	100.0 ± 0.00
Coverage Score	15.8 ± 3.40
Feature Accuracy	61.4 ± 0.66
Novelty Score	0.7 ± 0.02

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	40.365	-4.382	9.8%	OK
TB	3.299	1.133	-2.165	65.6%	Drift
DB	1.486	0.392	-1.094	73.6%	Drift
Alkphos	290.576	206.255	-84.321	29.0%	Drift
Sgpt	80.714	31.495	-49.219	61.0%	Drift
Sgot	109.911	35.458	-74.452	67.7%	Drift
TP	6.483	6.447	-0.036	0.6%	OK
ALB	3.142	3.320	+0.178	5.7%	OK
A/G Ratio	0.947	1.041	+0.094	9.9%	OK
Selector	1.286	2.000	+0.714	55.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	76.8 ± 0.60
Correlation Stability (*2)	88.8 ± 0.87
Distribution Consistency (*3)	87.4 ± 0.66
Entropy Preservation (*4)	59.0 ± 0.89
Mean Z Score Drift (*5)	0.4 ± 0.01
Categorical Fidelity (*6)	91.5 ± 5.00
Privacy Risk (*7)	36.7 ± 4.03
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	15.8 ± 3.40

Feature Accuracy (*10)	61.4 ± 0.66
Novelty Score (*11)	0.7 ± 0.02

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	N/A	0.6470085470085473	-0.027350427350427343	
F1	0.7821343746683586	N/A	0.7219976719216055	-0.06013670274675326	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	N/A	0.6692307692307693	-0.03760683760683765	
F1	0.8113180488818635	N/A	0.7496272814158431	-0.06169076746602046	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6803418803418804	N/A	0.6666666666666667	-0.013675213675213682	
F1	0.7872041050336864	N/A	0.7573451591423772	-0.029858945891309184	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6965811965811965	N/A	0.6358974358974366	-0.06068376068376074	*
F1	0.8057756214374322	N/A	0.6931575588639738	-0.11261806257345845	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	N/A	0.6470085470085473	-0.05897435897435895	*
F1	0.8128002697439015	N/A	0.7073633737874203	-0.10543689595648109	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.706837606837606 8	N/A	0.579487179487179 4	-0.12735042735042734	*
F1	0.827909162452953 7	N/A	0.611401457899808 4	-0.21650770455314539	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.702564102564102 5	N/A	0.688888888888888 9	-0.01367521367521368 2	
F1	0.799790357154994 8	N/A	0.779745526186411 7	-0.02004483096858302 6	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	N/A	0.564102564102564 1	0.00512820512820512 1	
F1	0.558743211938091 7	N/A	0.568363303358180 9	0.00962009142008919 3	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.657264957264957 4	N/A	0.622222222222222 1	-0.03504273504273503	
F1	0.760217537042212 7	N/A	0.707410477438502 6	-0.05280705960371022 4	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.686324786324786 3	N/A	0.635042735042735	-0.05128205128205128	*
F1	0.797129931976849 3	N/A	0.699608086753457 3	-0.0975218452233921	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	54	Male	1.62251873947054 1	0.87748126052945 93	180	47	64	6.36488756317675 5	3.80992504211783 7	1.47748126052945 92
2	69	Male	1.38453629017538 55	0.58711357514615 46	146	12	24	6.19226814508769 3	3.78453629017538 5	1.56969086011692 37
3	47	Female	1.24095630651152 8	0.37638252260461 125	205	30	26	6.37787963832622	3.27713108046541 55	1.05291475678138 35
4	43	Male	0.78920849629144 61	0.2	192	27	19	6.15108105191975 6	3.04028954821120 2	0.94316601483421 59
5	36	Male	1.59339718649612 92	0.62479839228350 23	372	131	80	7.54130542604318	4.00660281350387 1	1.01570060289368 68
6	32	Male	0.84402234501374 9	0.22201117250687 447	157	27	21	7.65787710242438 1	4.36793296495875 35	1.3
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	23	Male	0.81711110566154 9	0.2	184	34	53	6.88555552830774 4	3.70533326793858 77	1.13688884529239 2
2	58	Male	0.78898930232901 57	0.19449465116450 781	190	20	24	6.01101069767098 5	2.96926872092422 26	0.95825802325323 83
3	39	Male	0.92936948382104 49	0.22936948382104 488	181	36	90	7.01189154853686 5	3.75873896764208 97	1.24964061247888 17
4	39	Male	0.84614849955762 16	0.2	191	27	20	6.18459399823048 6	3.22303949690335 05	1.03844549867286 46
5	42	Female	1.00278513815825 9	0.36222811052660 72	156	22	21	7.22724135921147 3	3.46501324868486 64	0.88111405526330 37
6	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Female	0.9	0.2	199	32	25	7.56107715853808 85	3.91676852438573 3	1.10379424389842 94
2	35	Male	1.91854309983092 27	1.03890732487319 22	227	34	132	6.09503305019725 7	2.80860920022543 7	0.83937083508172 07
3	40	Male	0.88074691416349 4	0.22049794277566 276	153	58	21	6.91925308583650 5	4.01925308583650 6	1.33975102861216 85
4	62	Male	1.81729909709078 36	0.80864954854539 17	265	36	38	5.66946148422713 1	2.56513670995443 55	0.80432477427269 59
5	45	Male	1.41231099843938 98	0.59450785602813 56	184	58	95	6.26988585914935 65	3.66439371517749 15	1.58494292957467 8
6	37	Male	1.19539956649892 45	0.32659981421382 48	167	19	20	6.60639925685529 9	3.21099969035637 5	0.90843998761425 5
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	1.67801983679495 41	0.66783080523315 26	285	85	41	7.47292532101405 4	4.0050945157809	1.09490548421909 93
2	49	Male	0.70040248889196 39	0.10034499047882 617	148	13	12	5.39971250793431 2	2.79982750476058 67	1.0
3	19	Female	0.58898636140256 27	0.18898636140256 267	202	13	13	6.21013638597437 3	3.16520457896156	0.98810910877949 88
4	28	Male	0.75531737940023 67	0.25177245980007 89	179	15	18	5.97658689700118 4	3.21772459800078 94	1.08268807429291 92
5	51	Male	0.63146004725668 83	0.13146004725668 828	197	12	13	7.37190070885032 4	3.58314042531019 44	0.93146004725668 83
6	59	Male	0.92021317448834 57	0.27340439149611 53	190	19	22	5.49255169354273 2	3.08616925952719 3	1.22659560850388 47
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	41	Male	0.9044520447857742	0.2659697124962783	113	21	22	7.027487380206782	3.0659697124962784	0.7274873802067826
2	58	Male	0.9380481318441797	0.2690240659220898 3	196	20	26	5.676096263688359	2.6929278022337306	0.8929278022337306
3	51	Male	0.62359962259349	0.11179981129674504	178	25	27	6.576698773428843	3.62359962259349	1.1882001887032505
4	25	Male	1.2830924842303836	0.4165674935066947	162	53	80	6.833560014841989	3.6000850055657466	1.0333049981447513
5	58	Male	1.5945410785365586	0.4315136928455 1955	203	42	26	4.835979531111233	2.2022329191328565	0.7456575381029877
6	41	Male	1.8734009300296335	0.9734009300296335	218	40	122	6.177238697958513	2.9785179539348 07	0.9265990699703666
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Male	1.3727430888567111	0.4147786583212666 5	163	17	22	7.470426833574 67	3.7840711389291113	1.0279470327930622
2	32	Male	3.659729470305721	1.5370834745608004	158	41	99	5.748479046623319	3.0958957415354798	1.3242395233116597
3	28	Male	1.1658134374406366	0.3863253749762547 4	180	63	38	6.079488062464382	3.2384641873931 46	1.0726507499525095
4	28	Female	0.8672535397879659	0.2336267698939829 6	137	28	26	6.930985840848137	4.0973590709541 54	1.3991196903180512
5	60	Female	0.8	0.2	162	18	18	6.31101471799444	2.8293725813185064	0.8146862906592534
6	36	Female	0.8	0.2	141	22	25	6.4949950987101905	3.5668123893847 71	1.1987706330852634
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Male	0.7543395333380588	0.13622635555870594	162	51	46	5.544150377807706	2.7173581333522354	0.9181131777793533
2	35	Male	0.8131383769410031	0.2	183	30	32	6.4	3.773723246117994	1.3737232461179938
3	58	Male	0.7982900992422161	0.2	179	31	25	8.17948119090659	4.384610893179944	1.0982900992422161
4	38	Female	0.6228415574471091	0.1228415574471919	167	23	36	5.762950655317345	2.785792212764454	0.8771584425528909
5	40	Male	0.9	0.2	200	30	23	7.594495251946251	3.799082541991041	1.0009174580089581
6	46	Female	0.9195839885617043	0.2829866571347357	150	14	14	6.351530617100719	3.239167977123408	1.0
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	35	Male	0.9523746999181057	0.3103122499317549	163	50	41	5.368249799945404	2.5	0.8579375500136409
2	31	Male	0.7468468489132772	0.1468468489132772	142	25	24	6.897297266300842	3.737837813040673	1.1735135079341517
3	16	Male	0.8739456961735292	0.2369728480867645 8	312	67	71	6.995782784694117	3.666377816390441	1.1
4	59	Male	0.8658214459366291	0.2329107229683145 5	190	25	31	6.340517709538653	3.235446385158427	0.9987321689049437
5	51	Male	0.8	0.2	230	12	18	6.376136399550791	3.115909066966139	0.9974999736627532
6	27	Male	1.4215462073201437	0.6694549756115106	105	25	51	8.413181280485611	4.747908768291367	0.9694167958929313
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	52	Male	0.8512933725682852	0.2504311241894284	201	18	28	5.205173490273141	2.3521556209471424	0.8
2	19	Male	0.8487895219242545	0.2	289	48	42	7.607262868454473	3.8975790438485087	1.0463685657727635
3	32	Male	0.7796522920910348	0.1898261460455174	135	29	27	6.966001359046899	4.145653651137934	1.4084353144096564
4	30	Male	0.7099874463516327	0.2	193	32	35	7.490012553648367	3.65493095493398	0.9639476394718 42
5	46	Male	1.7919965320026607	0.7946643546684405	110	30	18	6.844219717557477	3.695553628890367	1.0991107257780734
6	35	Male	0.7908470423749884	0.1908470423749883	158	31	39	5.926776338999907	2.2274588728750353	0.5366118305000468
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	56	Male	1.2980994969616009	0.3326998323205635	206	39	28	6.18643562495267	2.9268678963160664	0.8115334451196424
2	18	Female	0.8156214391529677	0.2	198	34	31	6.640592952376709	3.5937286349178066	1.166248575661187
3	52	Male	0.7101801213498666	0.1550900606749333	165	45	110	5.423355576411867	2.958085394387066	1.306587727531
4	54	Male	1.4020192010477952	0.6989903994761023	195	36	16	7.879807989522046	3.6899039947610 3	0.8989903994761024
5	39	Male	0.9439941235202464	0.2439941235202638	207	26	24	7.46801762943926	4.063964741121478	1.2199706176012 32
6	65	Male	0.7	0.2	162	24	20	6.442950524199951	3.2214752620999754	1.0
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3